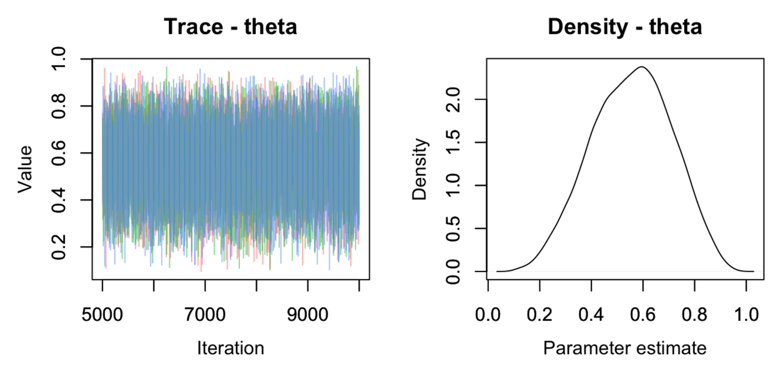
```
##           mean        sd      2.5%      50%     97.5% Rhat n.eff
## mu    5.141125 0.9365031 3.2300139 5.148315  7.028937    1 14796
## nu    8.643353 7.0519522 1.0996277 6.612103 27.382071    1  6265
## sigma 2.165768 0.9703596 0.8715478 1.976407  4.613030    1  6374
```

Bernoulli, one group

```
flips <- c(1,1,1,1,0,0,0)
## number of observations
N <- length(flips)

## All data into one list
myData <- list(y = flips, N = N)

n.adapt = 1000
n.update = 10000
n.iter = 10000

### Just for repeatability
set.seed(1)

## We give the procedure a starting point for each parameter
initial_start <- list( theta = 0.5)

## Create computer object and adapt its sampling procedures
jagsModel_b <- jags.model(file="jags_bern.R", data=myData, inits=initial_start,
                          n.adapt=n.adapt, n.chains=3)
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 7
##    Unobserved stochastic nodes: 1
##    Total graph size: 10
##
## Initializing model
```

```
## Warm up
update(jagsModel_b, n.iter=n.update)

## sample
codaSamples_b <- coda.samples(jagsModel_b,
                            variable.names=c("theta"),
                            n.iter=n.iter, n.thin = 1)
MCMCtrace(codaSamples_b, pdf=FALSE)
```



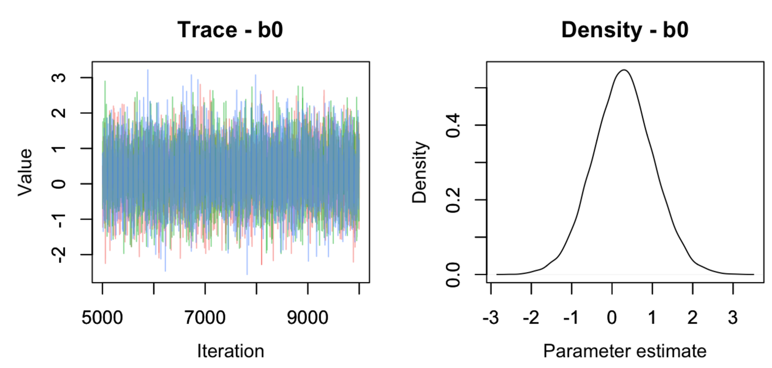

```
MCMCsummary(codaSamples_b)
```

```
##            mean        sd      2.5%       50%     97.5% Rhat n.eff
## theta 0.5551795 0.1559487 0.2478539 0.5603438 0.8408986    1 28813
```

```
## We give the procedure a starting point for each parameter
initial_start <- list( b0 = 0.5)

## Create computer object and adapt its sampling procedures
jagsModel_b <- jags.model(file="jags_bern2.R", data=myData, inits=initial_start,
                          n.adapt=n.adapt, n.chains=3)
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 7
##    Unobserved stochastic nodes: 1
##    Total graph size: 15
##
## Initializing model
```

```
## Warm up
update(jagsModel_b, n.iter=n.update)

## sample
codaSamples_b <- coda.samples(jagsModel_b,
                            variable.names=c("b0"),
                            n.iter=n.iter, n.thin = 1)
MCMCtrace(codaSamples_b, pdf=FALSE)
```




```
MCMCsummary(codaSamples_b)
```

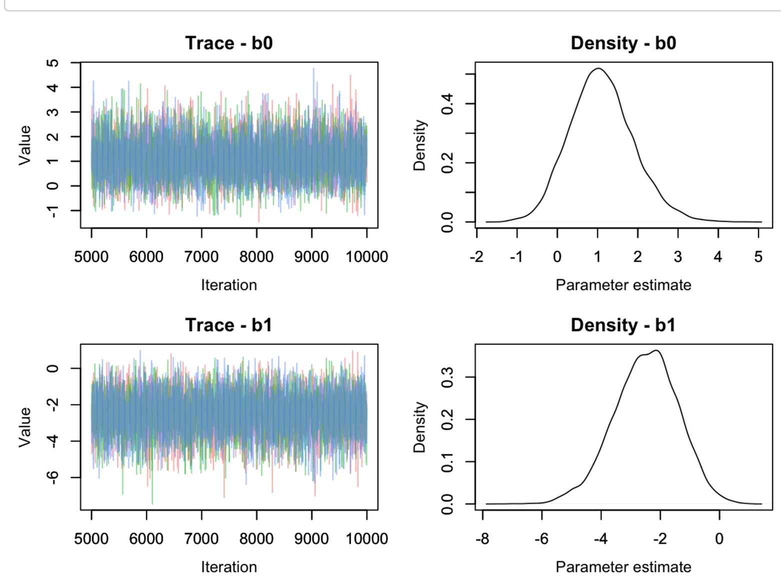
```
##         mean        sd      2.5%       50%    97.5% Rhat n.eff
## b0 0.2847718 0.7523531 -1.192674 0.2824394 1.778051    1 18071
```

Bernoulli, two groups

```
flips <- c(1,1,1,1,1,1,0,
           1,0,0,0,0,0,0)
## number of observations
N <- length(flips)
x <- c( rep(0,7), rep(1,7))

## All data into one list
myData <- list(y = flips, x=x, N = N)

n.adapt = 1000
n.update = 10000
n.iter = 10000

### Just for repeatability
set.seed(1)

## We give the procedure a starting point for each parameter
initial_start <- list( b0 = 1, b1 = -1)

## Create computer object and adapt its sampling procedures
jagsModel_b <- jags.model(file="jags_bern_2grps.R", data=myData, inits=initial_start,
                          n.adapt=n.adapt, n.chains=3)
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 14
##    Unobserved stochastic nodes: 2
##    Total graph size: 42
##
## Initializing model
```

```
## Warm up
update(jagsModel_b, n.iter=n.update)

## sample
codaSamples_b <- coda.samples(jagsModel_b,
                            variable.names=c("b0", "b1"),
                            n.iter=n.iter, n.thin = 1)
MCMCtrace(codaSamples_b, pdf=FALSE)
```




```
MCMCsummary(codaSamples_b)
```

```
##         mean        sd       2.5%       50%      97.5% Rhat n.eff
## b0  1.108157 0.7865483 -0.3481021  1.076995  2.7479261    1  9189
## b1 -2.499682 1.0818191 -4.7603828 -2.461772 -0.4854842    1  9101
```

```
df <- MCMCchains(codaSamples_b)
HPDinterval(as.mcmc(df[,1]-df[,2]))
```

```
##          lower    upper
## var1 0.4666805 6.975999
## attr(,"Probability")
## [1] 0.95
```